The Controllable Decomposition

Basic Idea of Decomposition

Let $\dot{x} = Ax + Bu$ but not c.c.

let $\dim(\mathrm{Im}(\mathcal{M})) = d < n$

- Define a subspace w/ everything but the vectors in $\mathcal{V}$

$$W = (\mathrm{Im}(\mathcal{M}))^{\perp} = \mathcal{V}^{\perp}$$

Use the Fundamental Theorem of Linear Algebra

$$\dim(W) = n - d$$

Furthermore, we can write $\mathbb{R}^n = \mathrm{Im}(\mathcal{M}) \oplus W$

set addition

Decomposition Example

$$\dot{x} = \begin{bmatrix} 2 & 0 & 0 \\ 2 & 2 & 2 \\ 3 & 0 & 1 \end{bmatrix} x + \begin{bmatrix} 1 \\ -2 \\ 1 \end{bmatrix} u \qquad \mathcal{M} = [B \quad AB \quad A^2B]$$

$\mathrm{rank}(\mathcal{M}) = 2 \Rightarrow$ Not completely controllable

What is $\mathrm{Im}(\mathcal{M})$?

Returns an orthonormal basis

- Use Matlab's "orth" command to find a basis set for $\mathcal{M}$

$$\mathrm{orth}(\mathcal{M}) = \begin{bmatrix} 0.4204 & -0.5685 \\ 0.8040 & 0.5946 \\ 0.4204 & -0.5685 \end{bmatrix}$$

v_1 v_2

v_1 & v_2 are a basis set for $\mathrm{Im}(\mathcal{M})$

$\Rightarrow \mathrm{Im}(\mathcal{M}) = \mathrm{span}\{v_1 \; v_2\}$

- What about $W = (\mathrm{Im}(\mathcal{M}))^{\perp} = N(\mathcal{M}^T)$

$$\mathrm{null}(\mathcal{M}^T) = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

w_1

$W = \mathrm{span}\{w_1\}$

w_1 is a basis set for $(\mathrm{Im}(\mathcal{M}))^{\perp}$

Write a basis of the image and nullspace

$$\mathrm{Im}(\mathcal{M}) = \mathrm{span}\{v_1\ v_2\} \qquad W = \mathrm{span}\{w_1\}$$

Make a change of coordinates

$$T = [v_1\ v_2\ w_1]$$

$$T\hat{x} = x \quad \text{or} \quad \hat{x} = T^{-1}x$$

$$\text{let } \hat{x} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$$

$$\Rightarrow T\hat{x} = \begin{bmatrix} 0.4204 \\ 0.8040 \\ 0.4204 \end{bmatrix} \in \mathrm{Im}(\mathcal{M})$$

$$\text{let } \hat{x} = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix} \Rightarrow T\hat{x} \in \mathrm{Im}(\mathcal{M})$$

i.e. First two elements are states we can control

Write out a new system

$$\dot{\hat{x}} = \hat{A}\hat{x} + \hat{B}u$$

$$\hat{A} = T^{-1}AT \qquad \hat{B} = T^{-1}B$$

Looking at the example above:

$$\hat{A} = \left[\begin{array}{cc|c} 3.3132 & -0.7466 & 1.0301 \\ \hline 2.31 & 0.6868 & -3.465 \\ 0 & 0 & -1 \end{array}\right] \qquad \hat{B} = \begin{bmatrix} -0.3132 \\ -2.31 \\ 0 \end{bmatrix}$$

The point of this is the form of the matrices:

$$\hat{A} = \begin{bmatrix} \underset{2\times 2}{\hat{A}_{11}} & \underset{2\times 1}{\hat{A}_{12}} \\ \underset{1\times 2}{0} & \underset{1\times 1}{\hat{A}_{22}} \end{bmatrix} \qquad \hat{B} = \begin{bmatrix} \underset{2\times 1}{\tilde{B}_1} \\ \underset{1\times 1}{0} \end{bmatrix}$$

This is the controllable variables at the top and the uncontrollable at the bottom

Cool Fact:

$$\text{rank}([\hat{B} \;\; \hat{A}_{11}\hat{B}_1]) = 2 = \text{rank}(\mathcal{M})$$

We have a sub-system defined by $(\hat{A}_{11}, \hat{B}_1)$ that is completely controllable

Decomposition

What we have shown is that we have taken the system to a basis / decomposition where the first part is C.C. and the second part is not.

Fact: Iff the system is stabilizeable then the $\text{eig}(\hat{A}_{22})$ are in the Left Half plane

I) General Form

$$\mathbb{R}^n = \underbrace{\text{Im}(\mathcal{M})}_{\substack{\dim = d \\ \mathcal{V}}} \overset{\text{set plus}}{\oplus} \underbrace{(\text{Im}(\mathcal{M}))^{\perp}}_{\substack{\dim = n-d \\ \mathcal{W}}}$$

$$\Rightarrow \quad \mathcal{V} = \text{Im}(\mathcal{M}) \quad \text{and} \quad \mathcal{W} = (\text{Im}(\mathcal{M}))^{\perp}$$

Let $v_1, \cdots, v_d$ be a basis set for $\mathcal{V}$, and let $w_1 \cdots w_{n-d}$ be a subspace for $\mathcal{W}$

A) Define Transform

$$T = [v_1 \cdots v_d \; w_1 \cdots w_d] \in \mathbb{R}^{n \times n} \rightarrow \begin{array}{l} \hat{A} = T^{-1}AT \\ \hat{B} = T^{-1}B \end{array}$$

We get the state definition

$$\hat{x} = \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} \qquad \hat{x}_1 \in \mathbb{R}^d \qquad \hat{x}_2 \in \mathbb{R}^{n-d}$$

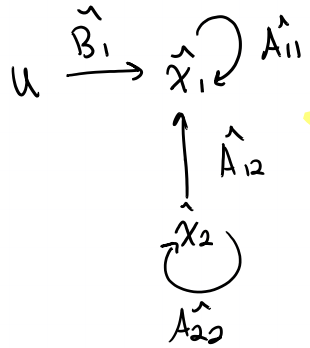
New states will have dynamics of

$$\begin{bmatrix} \dot{\hat{x}}_1 \\ \dot{\hat{x}}_2 \end{bmatrix} = \underbrace{\begin{bmatrix} \hat{A}_{11} & \hat{A}_{12} \\ 0 & \hat{A}_{22} \end{bmatrix}}_{\hat{A}} \begin{bmatrix} \hat{x}_1 \\ \hat{x}_2 \end{bmatrix} + \underbrace{\begin{bmatrix} \hat{B}_1 \\ 0 \end{bmatrix}}_{\hat{B}} u$$

Fact: $(\hat{A}_{11}, \hat{B}_1)$ is completely controllable

$\Rightarrow$ iff $\hat{x}_2$ $(\text{eig}(\hat{A}_{22}) < 0)$ then we can make $\hat{x}_1$ do what we want

Block Diagram Representation



u does not affect $\hat{x}_2$

$\xrightarrow{\hat{A}_{11}}$ Indicates "affects"

↳ Indicates "Through matrix $\hat{A}_{11}$"

State Feedback With Controllable Portion

$$\dot{\hat{x}} = \begin{bmatrix} \hat{A}_{11} & \hat{A}_{12} \\ 0 & \hat{A}_{22} \end{bmatrix} \hat{x} + \begin{bmatrix} \hat{B}_1 \\ 0 \end{bmatrix} u$$

$\hat{A}_{11} \in \mathbb{R}^{d \times d}$ $\quad \hat{A}_{22} \in \mathbb{R}^{(n-d)\times(n-d)}$

$\hat{A}_{12} \in \mathbb{R}^{d \times (n-d)}$

$(\hat{A}_{11}, \hat{B}_1)$ is completely controllable

I) Let input be defined as

$u = -k\hat{x}$ $\quad k = [k_1 \;\; k_2]$ $\quad k \in \mathbb{R}^{n \times m}$

$k_1 \in \mathbb{R}^{n \times d}$ $\quad k_2 \in \mathbb{R}^{n \times (n-d)}$

A) Closed Loop Dynamics

$$\dot{\hat{x}} = \begin{bmatrix} \hat{A}_{11} & \hat{A}_{12} \\ 0 & \hat{A}_{22} \end{bmatrix} \hat{x} + \begin{bmatrix} \hat{B}_1 \\ 0 \end{bmatrix} [k_1 \;\; k_2] \hat{x}$$

$$\Rightarrow \begin{bmatrix} \hat{A}_{11} - \hat{B}_1 k_1 & \hat{A}_{12} - \hat{B}_1 k_2 \\ 0 & \hat{A}_{22} \end{bmatrix} \hat{x} = M\hat{x}$$

Fact: M is upper block-triangular and any such matrix eigenvalues given are the eigenvalues of the triangular blocks.

$$\text{eig}(M) = \text{eig}\{(\hat{A}_{11} - B_1 k_1) ; \text{eig}(A_{22})\}$$